

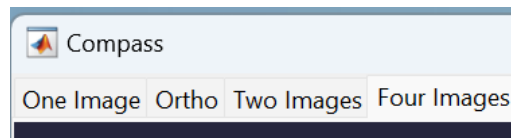
1. Starting Compass

To start Compass, first navigate to the Compass folder in Matlab. Alternatively the Compass folder can be added to the Matlab path (and saved) to avoid this step in the future.

There are three Compass ‘modes’: *CompassFull*, *CompassImageAnlz*, and *CompassScripts*. Type any of these modes in the Matlab Command Window and hit enter to start the software.

2. Compass Viewports:

The *CompassFull* mode has 4 different viewport options: ‘One Image’, ‘Ortho’, ‘Two Images’, and ‘Four Images’. Tabs to access these different viewports are located to the top left of the software. There are also 4 ‘Script’ tabs which are not covered in this document. The *CompassImageAnlz* mode only has the **Ortho** and **Two Images** tabs. These are the tabs primarily used for the image analysis discussed below.



2.1. Common Viewport Components

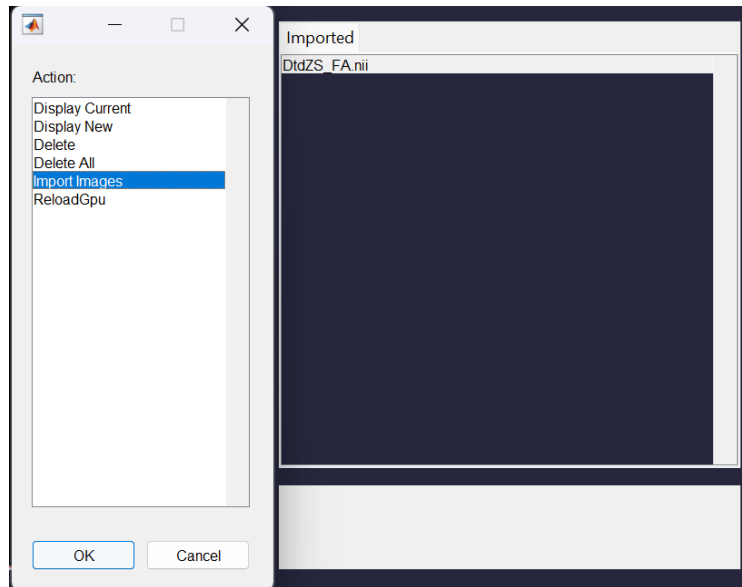
At the top right of each viewport are ‘Script’, ‘General’, ‘Analysis’, and ‘Info’ tabs.

- The **Script** tab is not covered in the document.
- The **General** tab contains some simple image contrast and navigation options.
- The **Analysis** tab provides information pertain to Region of Interest (ROI) analysis.
- The **Info** tab lists image information (if there is any to list).

The **Imported** tab at the bottom right is used to load images into Compass. It is also used to select which images to display in the viewport.

2.2. Importing Images

To import an image into Compass, drag an image from the file explorer into the **Imported** tab, (or into the blue square area under the **Imported** tab). Once imported, the file name will be listed within the **Imported** tab (as shown for the image 'DtdZS_FA.nii' below).



Alternatively, a right-click within the **Imported** tab space will open up the selection box shown above (to the left of the **Imported** tab). The selection 'Import Images' will open File Explorer for the user to choose the image of their choice.

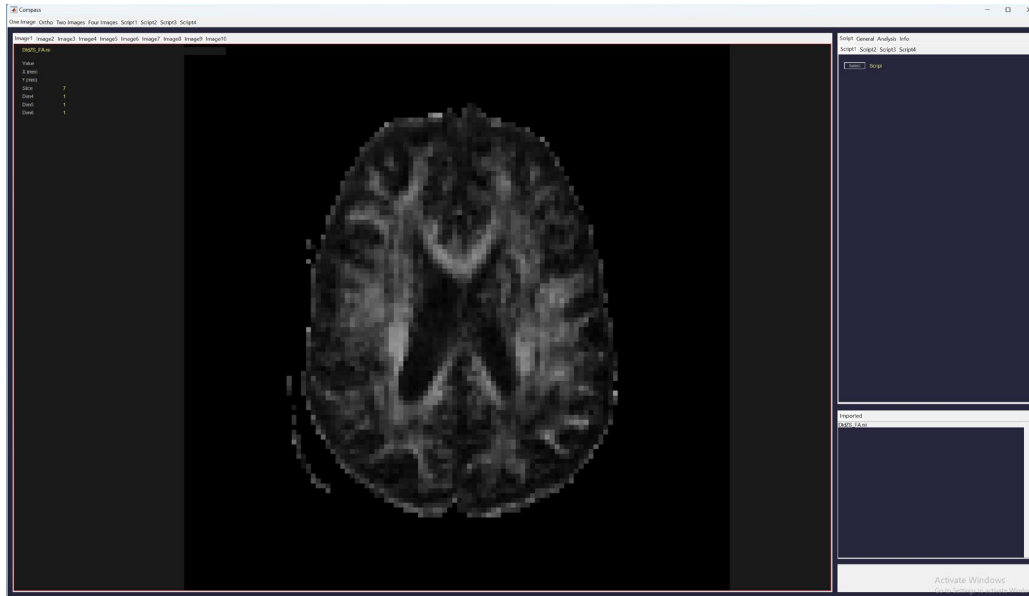
2.3. Displaying Images

Selected images within the **Imported** tab will be highlighted in blue. A right-click within the **Imported** tab space will open up the selection box with 'Display' options. These 'Display' options will vary depending on the Viewport used.

The **Ortho** and **Two Image** viewports, also have **Load** buttons within the image display area. Clicking the Load button will display the image selected in the **Import** tab.

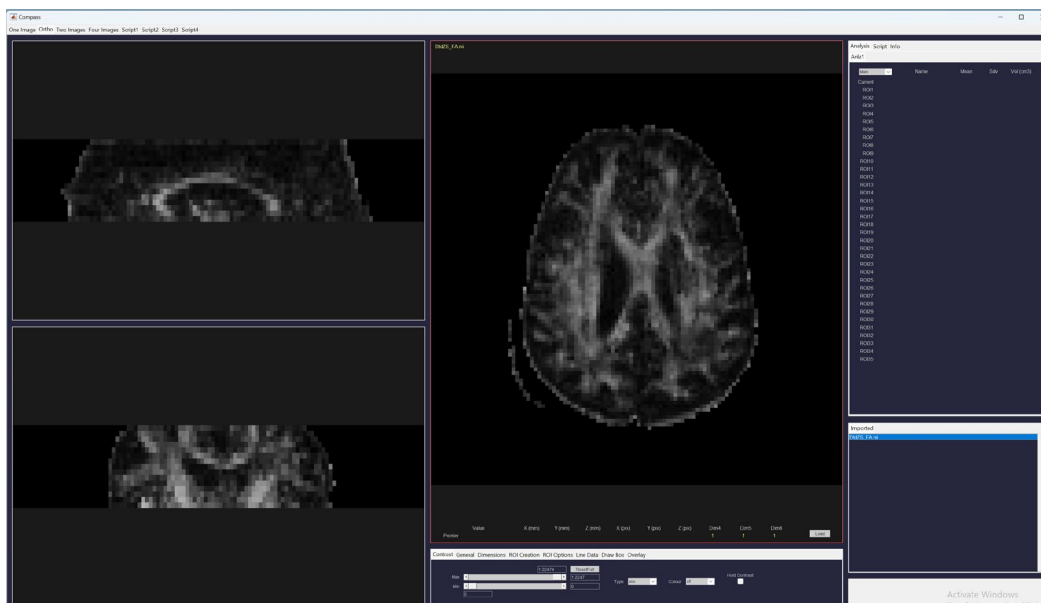
2.4. One Image Viewport

The **One Image** viewport was created as an interface for the 4.7T Varian Inova scanner. However, it does serve to view images in a large format. Pixel intensities for mouse locations are given at the top left. The **General** tab (top right) gives some simple functionality for image contrast and navigation.



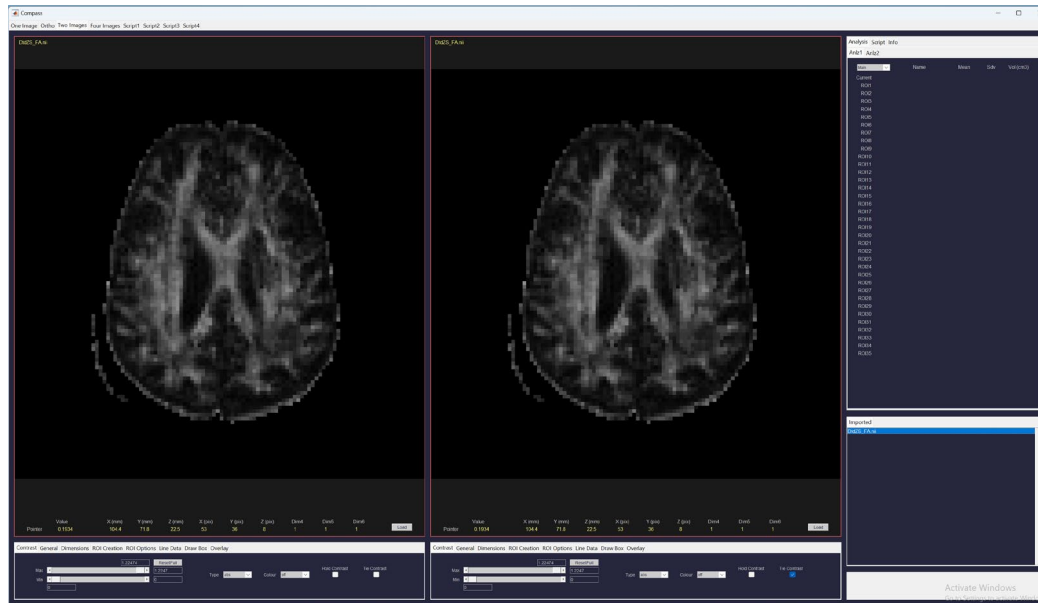
2.5. Ortho Viewport

The **Ortho** viewport was created to be the primary analysis viewport. Pixel intensities of mouse locations are given in the gray space below the largest figure on the right. These pixel intensities are displayed for the orientation selected (by mouse click). Note that the selected orientation is highlighted by a red box.



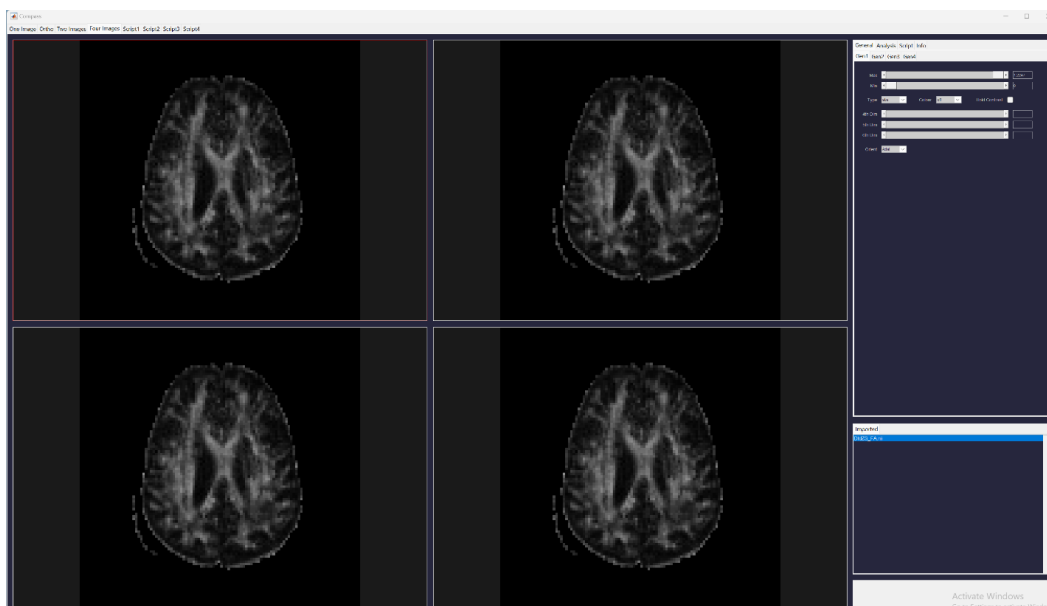
2.6. Two Images Viewport

The **Two Images** viewport was created for side-by-side image comparison. These images can be ‘tied’ together to simultaneously scroll through slices of both images.



2.7. Four Images Viewport

The **Four Images** viewport was also created for side-by-side image comparison. However, the this viewport has less analysis functionality than the **Two Images** viewport.



3. Image Navigation:

3.1. Slice Change

The mouse scroll wheel navigates through the out-of-plane dimension ('slices') of the selected image. The current out-of-plane dimension number will be updated accordingly.

3.2. Dimension Change

If the images has more than 1 dimension, the **Dimensions** tab can be used to navigate through the 4th, 5th and 6th dimensions.

Shortcut: Press and hold the number of the dimension to be changed and use the mouse scroll wheel to navigate through that dimension.

3.3. Orientation Change

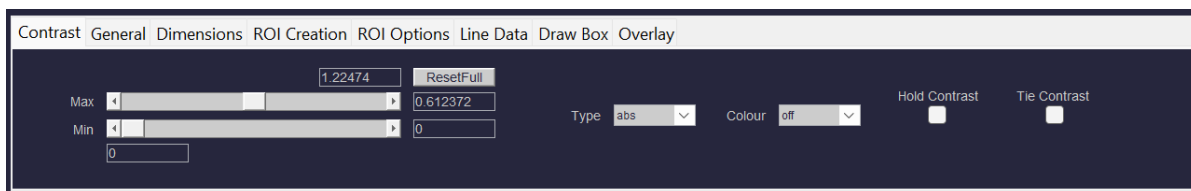
The orientation of the image can be changed on the **Dimensions** tab (Axial, Sagittal, Coronal).

4. Image Zooming:

Press and hold the Shift-key and use the mouse scroll-wheel. This will zoom in (or out) on the location of the mouse in the image. Note that you can zoom at any time.

5. Image Contrast:

Maximum and minimum image intensity values can be adjusted with the associated sliders in the **Contrast** tab. Alternatively, maximum and minimum values can be entered manually in the boxes to the right of these sliders. Images are originally displayed such that the largest pixel value in the image is assigned to an image intensity value of 1 (or full intensity). If a different value is entered in the box to the right of the max-slider, that number will be assigned the image intensity value of 1. The number above the max-slider can be altered to change the range of the sliders. Clicking **ResetFull** will reset the contrast to the to originally displayed levels.



Hold Contrast will maintain the specified contrast when a new image is loaded.

Tie Contrast (in the Two Images viewport) will connect the contrast slider bars of the two images.

Shortcuts:

- Press and hold the ~ (**tilde**) key and use the mouse stoll wheel to change Max.
- Press and hold the **Ctrl** key and use the mouse stoll wheel to change Min.
- Press the **p** key toggle between colour and non-colour images.

6. Image Tying (Two Images Viewport):

If the images in panel 1 and panel 2 are the same size, navigations through these images can be tied together in the **General** tab. Tying the cursor will add an 'x' to the other image panel, however, tying the cursor can be problematic for ROI drawing features (below).

7. Line Analysis (Intensity Profile):

Use the tab **Line Data** to plot image intensity along a line through an image. First click **Activate** to start drawing a line. The cursor will become a cross. Left-click in the image to start a line, and right-click to finish a line. Once a line is finished, its length and angles will be listed. The tool will remain activated to draw a second (and third) line. Only 3 lines can be drawn at a time. Lines are removed with the **Delete** button. The **Plot** button will display the image intensity profile along the drawn line. If two images are tied (Two Images viewport), two lines will be drawn as well as 2 intensity profiles.

Shortcut: Pressing **l** will activate the line drawing tool

8. Draw Box

This feature is currently not supported.

9. Overlay

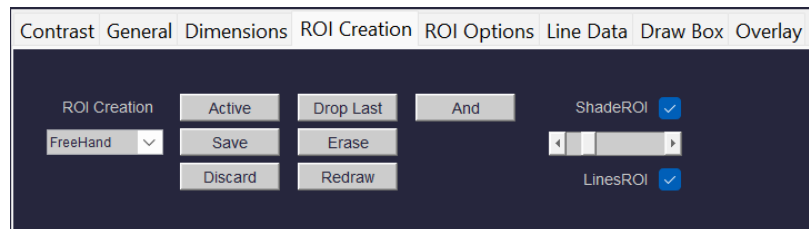
To plot another image as an overlay on the current image, first select the overlay image in the **Imported** tab, and then click Load in the **Overlay** tab. The two images must have the same dimensions. Colour and transparency can be adjusted with the associated selector and slider.

The numbers to the right of the slide are for the overlay contrast, but these are not currently supported.

The square box to the far right offers a quick way to select different overlays (with partial transparency).

10. ROI Creation

The **ROI Creation** tab offers several ways to draw and edit an ROI. Note that the info bars in the bottom right corner of Compass will display instructions for active tools.



10.1. Activating ROI Creation

Click **Activate** to begin creating an ROI. The cursor will become a cross (or modified cross, depending on the ROI drawing method). Different ROI creation methods can be selected in the drop-down menu to the left. These will be discussed in Section 11.

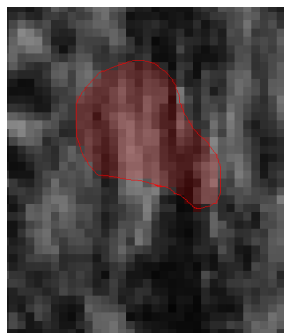
Shortcut: Pressing **z** will activate ROI creation

10.2. ROI Display

The current ROI being constructed will be shown in **red**. If **ShadeROI** is selected, the ROI will be shaded according to the transparency level of the slider bar. If **LinesROI** is de-selected, the line around the ROI will be removed. An example ROI is shown below.

Shortcut:

- Pressing **s** will toggle ROI shading
- Pressing **a** will toggle the ROI line display



ROI values of mean image intensity, standard deviation, and volume will be given the **Analysis** tab (top right of Compass). These values will be shown in red beside the *Current* heading.

10.3. ROI - Multiple Region Composition (Drop Last)

ROIs can be composed of multiple regions. As long as the ROI tool is active (**Active** button green), any new region drawn will be added to the ROI. New regions may be on the same slice, or different slices. If a new region is not acceptable, that region can be removed by clicking **Drop Last**. This will remove the last drawn region.

Shortcut: Pressing **x** will drop the last region drawn.

10.4. ROI - Erase

Activating the erase function (**Erase** button red) allows the user to erase portions of the ROI. As long as the **Erase** button is active any region drawn will function to *erase* (and not add). Re-clicking the activated **Erase** button will de-activate it. Note that **Drop Last** also works for 'erase regions'.

Shortcut: Pressing **e** will toggle the ROI erase function

10.5. ROI - Redraw

If the erase region includes areas that are not part of the ROI, this will not affect the ROI (it is *always* accurately represented the shaded region), but the extraneous lines can be distracting/confusing. The **Redraw** function can be used to properly outline the ROI. Activating the redraw function (**Redraw** button red) will turn the shaded ROI blue, and the cursor will become a modified cross. The redraw function 'initiates the ROI to be rebuilt'. Each blue shaded region clicked with the modified cross will be added to the 'new' ROI. Blue shaded regions not clicked will be excluded from the 'new' ROI. The ROI lines will be redrawn to match the shaded region. Re-clicking the **Redraw** button will complete the redrawing process. Re-clicking **Redraw** without selecting a blue region will exit the redraw function without redrawing.

10.6. ROI – And Function

Activating the 'and' function (**And** button red) will turn the shaded ROI green. Any new region draw will be 'anded' with the shaded ROI in green. The resultant ROI will now just be the portion common to both regions. Note that the resultant ROI is the shaded red (or brown in this case) portion. As long as the **And** button is active, any region drawn will function to *and* (and not add). Re-clicking **And** without drawing any region will exit the and function and return the original ROI. Otherwise, re-clicking the **And** button will complete

the ‘anding’ process, and automatically initiate the redraw function (**Redraw** button **red**). Select the shaded blue regions as describe above. When finished, deactivate the redraw function. Note that **Drop Last** also works for ‘and regions’.

10.7. ROI – Shifting

Any active (**red**) ROI can be shifted up, down, left, right, in (plane), out (of plane). This can only be done using the short cuts below.

Note: If a region has ‘holes’ not defined by a line (possible cases: seeded ROI, or redrawn ROI), these ‘holes’ will be lost by an ROI shift.

Shortcuts:

- Pressing **i** will shift the ROI up
- Pressing **m** will shift the ROI down
- Pressing **j** will shift the ROI left
- Pressing **k** will shift the ROI right
- Pressing **,** will shift the ROI back (slice dimension)
- Pressing **.** will shift the ROI forward (slice dimension)

10.8. ROI – Discarding

An ROI can be discarded with the **Discard** button. This will also deactivate the ROI creation tool.

Shortcut: Pressing **v** will discard an ROI and deactivate the ROI creation tool.

10.9. ROI – Saving (within Compass)

To save an ROI, click the **Save** button in the **ROI Creation** tab. You will be prompted to enter a name for your ROI. After doing so, your ROI will change colour (**cyan**, if it is the first ROI you have saved). It will now be listed as **ROI1** in the **Analysis** tab. The yellow label mean the ROI is selected and visible. Clicking the yellow ROI label will de-select the ROI and make it in-visible. Note that ‘saving’ an ROI in this context just saves the ROI within Compass. Saving the ROI to a file will be described below.

Note that once an ROI has been saved, it can still be edited in the future. If an ROI is to contain may ‘parts’ or ‘steps’ it is good to save the ROI as it is being built. This will ensure that previous work is not lost if there is an error.

11. ROI Creation Methods

There are currently 7 ROI Creation methods which can be selected from the ROI Creation drop-down menu. These are described below.

For each drawing tool, the info area in the bottom right of Compass provides instruction for ROI creation.

Note: In the Ortho tab, ROIs can be drawn on any of the three orientations.

11.1. FreeHand

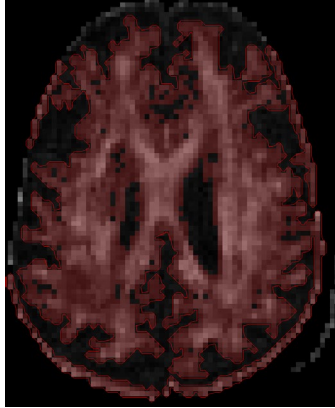
The cursor will be a cross when the FreeHand tool is active. Left click to start drawing a region, right click to finish drawing the region.

11.2. Seed

The cursor will be a modified cross when the Seed tool is active. Clicking a pixel location in the image with an intensity greater than the **Seed Value** listed to the right of the slider will seed an ROI around that point (if the drop-down to the right of the slider is 'Above'). The algorithm checks pixels to the right until it finds a pixel less than the seed. It then draws the boundary of the region in a counter-clockwise manner according to the **Seed Value**. Pixels in the region below the **Seed Value** are not included. If the drop-down menu to the right of the slider is 'Below', a region with values below the **Seed Value** will be created.

Note that sometimes the seeding algorithm will not produce the desired result. If there is an image 'hole' to the right of the mouse click (and the drop-down is 'Above'), the seed algorithm will circle that hole. Click **Drop Last** to delete this region (or press **x** as described above), and try seeding again from a different pixel location that does not have a hole to the right.

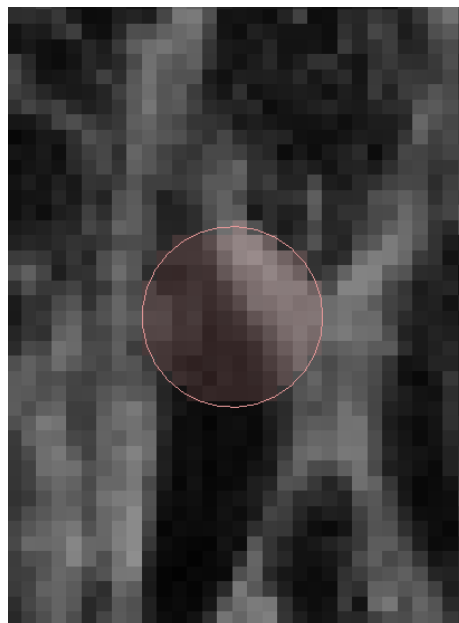




11.3. Sphere

The cursor will be a modified cross when the Sphere tool is active. Clicking anywhere on the image will create a (preliminary) spherical region with **Sphere Radius** listed (in pixels). This preliminary region will be shown in **pink**. The **Sphere Radius** can be modified using the number box. The **Sphere Centre** can also be modified using the number boxes (X,Y,Z). Alternatively, the centre of the sphere can be re-selected with the mouse (left click again), or can be shifted with the buttons to the right of the screen (not shown below). **Value** is the mean image intensity value in the spherical region, it will be updated with any changing or shifting of the sphere if **AutoUpdateROI** is selected in the **ROI Options** tab. A right mouse click anywhere will 'accept' the region and add it to the ROI (it will then turn **red**).

Sphere Radius	6	☒	Sphere Centre	51.5	49.9	8	☒
Value	0.26287						

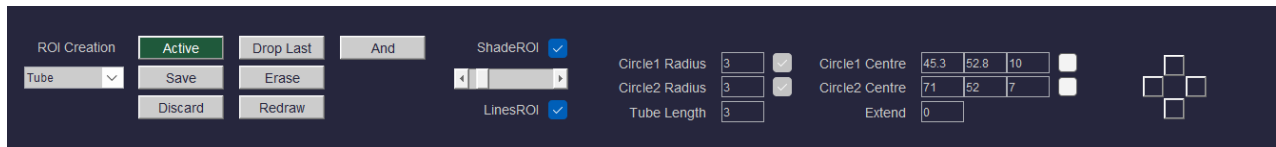


11.4. Circle

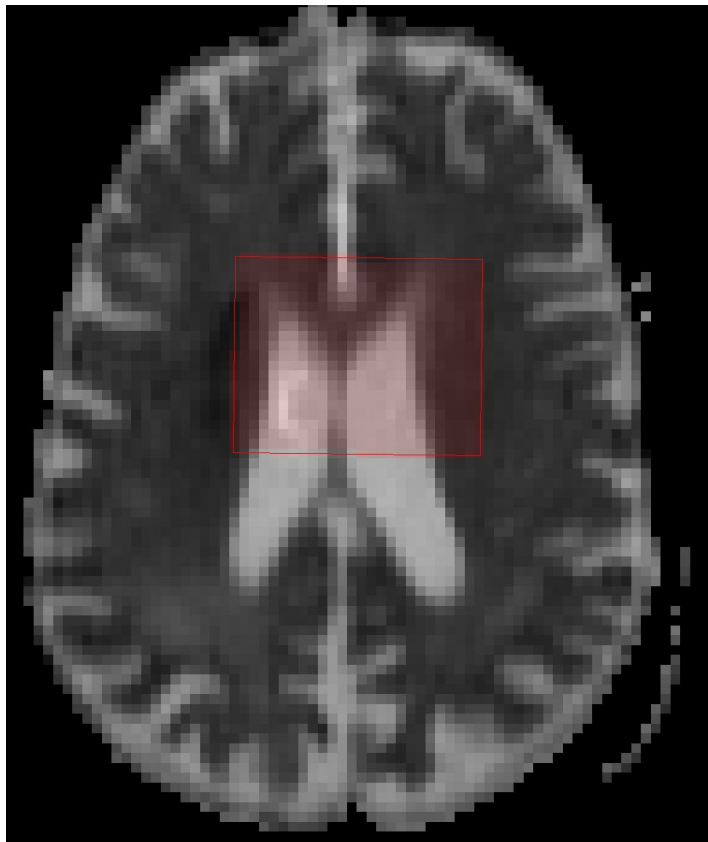
The Circle method is not currently supported.

11.5. Tube

The cursor will be a modified cross when the Tube tool is active. Clicking anywhere on the image will create a (preliminary) circle region with **Circle1 Radius** listed (in pixels).



Rect:



Box:



